Math Kangaroo Practice

Grade 3-4 · 20 questions · harder · seed 53324

Part A: Each question is worth 3 points

1. Five ribbons have these lengths.
Which ribbon is the longest?

A. 3800 mm ***B.*** 340 cm ***C.*** 4 m 50 cm ***D.*** 36 dm ***E.*** 43 dm

2. The odometer of a bike shows 32623 km. This number reads the same forwards and backwards.
After how many more kilometres will this happen again?

A. 11 ***B.*** 90 ***C.*** 100 ***D.*** 109 ***E.*** 110

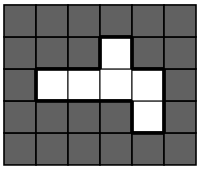
3. Anna, who is older than Lucy by 3 years and 3 days, was born on August 31, 2007.
When was Lucy born?

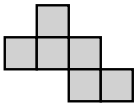
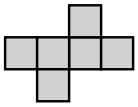
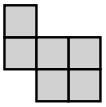
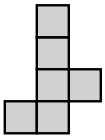
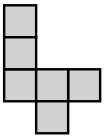
A. August 28, 2004 ***B.*** September 3, 2004 ***C.*** September 3, 2010
D. September 3, 2011 ***E.*** August 31, 2010

4. A big cube made of 8 × 8 × 8 = 512 small cubes is painted on the outside, also underneath, and then taken apart.
How many small cubes have exactly 3 painted faces?

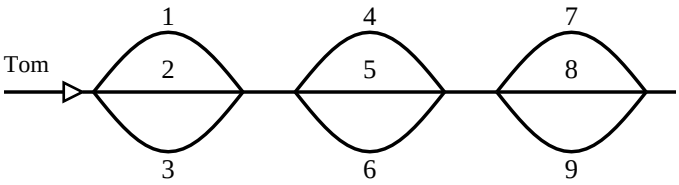
A. 5 ***B.*** 6 ***C.*** 7 ***D.*** 8 ***E.*** 24

5. Which piece covers the empty area exactly? The pieces may be turned and flipped over.



A.  ***B.***  ***C.***  ***D.***  ***E.*** 

6. Tom walks along the road from left to right and writes down every number on the way.

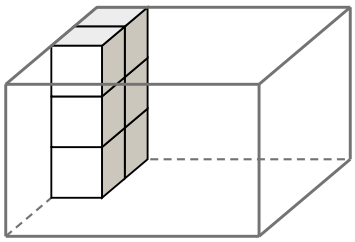


How many different groups of numbers can Tom write?

A. 6 ***B.*** 9 ***C.*** 13 ***D.*** 27 ***E.*** 29

Part B: Each question is worth 4 points

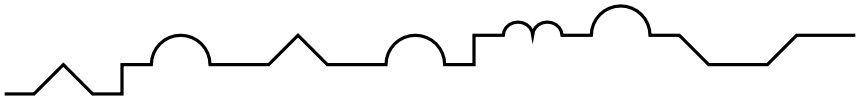
7. Sara's box is 5 cubes long, 4 cubes deep and 3 cubes tall. Some cubes are already inside, as in the picture. Every cube stands on the bottom or on another cube.



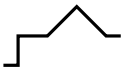
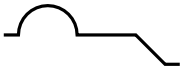
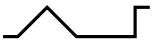
How many more cubes fit into the box?

A. 14 ***B.*** 44 ***C.*** 52 ***D.*** 53 ***E.*** 54

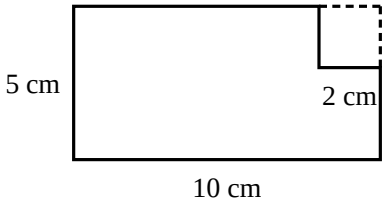
8. Lena cut a strip of paper along this line.



Which piece **can** come from the cut line?

A.  ***B.***  ***C.***  ***D.*** ***E.***

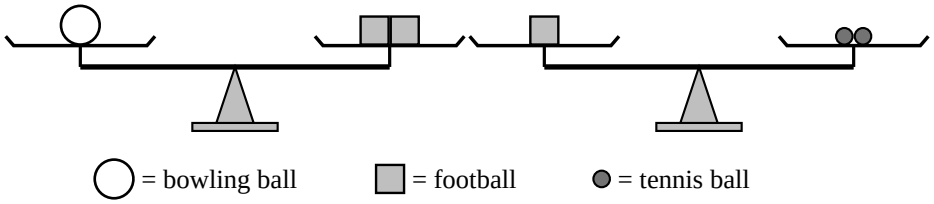
9. A rectangle is 10 cm long and 5 cm wide. A square with side 2 cm is cut off one corner.



What is the perimeter of the remaining shape?

A. 28 cm ***B.*** 30 cm ***C.*** 34 cm ***D.*** 36 cm ***E.*** 38 cm

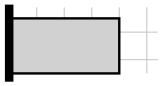
10. The two balances are level. One bowling ball weighs as much as 2 footballs, and one football weighs as much as 2 tennis balls.



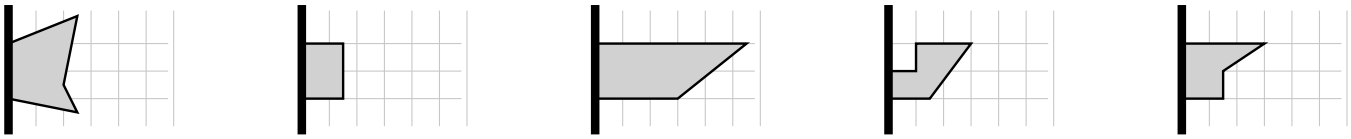
How many tennis balls weigh as much as 2 bowling balls?

A. 8 ***B.*** 9 ***C.*** 10 ***D.*** 11 ***E.*** 13

11. A rectangular flag flies on a pole in front of a wall, and the sun throws its shadow onto the wall. The wind turns and folds the flag, but the side sewn to the pole always stays on the pole. Tom draws the shadow at five different moments. Flat, the flag is 4 squares wide and 2 squares tall (see the picture). Some of the drawings are wrong.



the flag, flat



A B C D E

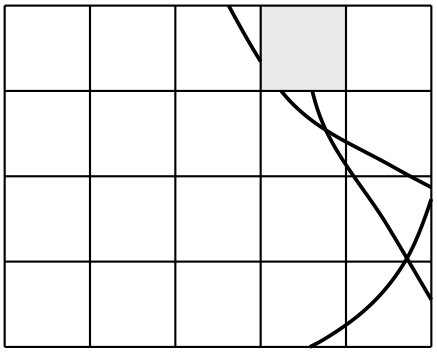
How many of the five drawings can be right?

A. 0 ***B.*** 1 ***C.*** 2 ***D.*** 3 ***E.*** 4

12. Aisha, Chloe, Omar, Ben and Nora each play a different instrument: the xylophone, the harp, the guitar, the violin and the drum. Omar uses a bow. Ben plays with sticks. Aisha doesn't play the harp. Chloe plays an instrument without strings. Chloe doesn't play the xylophone.
Which statement is true?

A. Nora plays the xylophone ***B.*** Aisha plays the violin ***C.*** Aisha plays the guitar
D. Chloe plays the harp ***E.*** Ben plays the drum

13. One square was cut out of this picture. Here it is five times, turned in different ways (and sometimes flipped over).



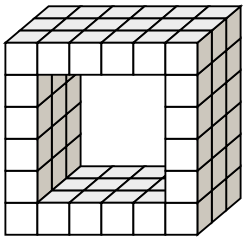
Which one fits back into the gap?

A. ***B.*** ***C.*** ***D.*** ***E.***

14. Only the even numbers from 2 to 120 are written down: 2, 4, 6, ..., 120.
How many digits are written?

A. 129 ***B.*** 127 ***C.*** 125 ***D.*** 120 ***E.*** 60

15. Nina built this tunnel out of identical cubes. The hole goes straight through from the front to the back. Nina wants to fill the hole completely, so that the tunnel becomes one solid block.



How many more cubes are needed?

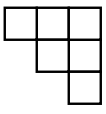
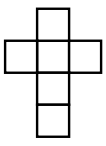
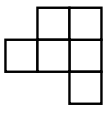
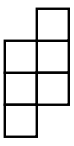
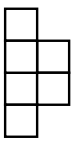
A. 12 ***B.*** 16 ***C.*** 32 ***D.*** 36 ***E.*** 48

Part C: Each question is worth 5 points

16. How many different letters are used to write the two words WINDOW and RAINBOW?

A. 6 ***B.*** 7 ***C.*** 8 ***D.*** 9 ***E.*** 12

17. Mia has two copies of the same piece. With which piece can she make a rectangle? Pieces may be turned and flipped over.

A.  ***B.***  ***C.***  ***D.***  ***E.*** 

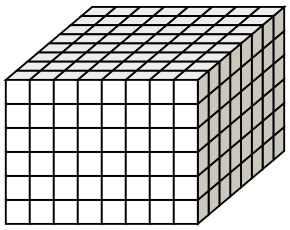
18. Five children, Grace, Emma, Ben, David and Iris, sit in a row of 5 chairs at the cinema. Grace and Emma are best friends and want to sit next to each other.
In how many different ways can the children sit?

A. 72 ***B.*** 48 ***C.*** 24 ***D.*** 8 ***E.*** 4

19. Lily has the cards 1, 2, 4 and 9. Using every card exactly once, with +, –, ×, ÷ and brackets, Lily wants to get the biggest possible number.
What is the biggest number Lily can get?

A. 126 ***B.*** 124 ***C.*** 120 ***D.*** 108 ***E.*** 16

20. Leo builds a brick 8 cubes long, 8 cubes deep and 6 cubes tall from red and blue cubes. All the cubes you can see from the outside are red. The cubes just below them are **also** red, so on every side (top, bottom, front, back, left and right) there are two layers of red cubes. All the other cubes, in the middle, are blue.



How many blue cubes are there?

A. 32 ***B.*** 30 ***C.*** 29 ***D.*** 24 ***E.*** 16

Answer Key

1.	**C**	**2.**	**C**	**3.**	**C**	**4.**	**D**	**5.**	**D**	**6.**	**D**
7.	**E**	**8.**	**E**	**9.**	**B**	**10.**	**A**	**11.**	**D**	**12.**	**C**
13.	**A**	**14.**	**B**	**15.**	**E**	**16.**	**C**	**17.**	**A**	**18.**	**B**
19.	**D**	**20.**	**A**								

Solutions

1. C (4 m 50 cm) — In cm: 3800 mm = 380 cm, 340 cm = 340 cm, 4 m 50 cm = 450 cm, 36 dm = 360 cm, 43 dm = 430 cm. The longest is 4 m 50 cm. [set-03/by_level/Grade_03-04/2003 Q10 · longest]

2. C (100) — The next such number is 32723, so 32723 – 32623 = 100. [set-03/by_level/Grade_03-04/2002 Q24 · next_palindrome]

3. C (September 3, 2010) — Older means born earlier, so Lucy was born 3 years and 3 days **after** August 31, 2007: 3 years later is August 31, 2010, then 3 more days: September 3, 2010. [set-03/by_level/Grade_03-04/2007 Q12 · younger_birthday]

4. D (8) — Only the corner cubes have 3 painted faces, and a cube always has 8 corners, however big it is. [set-03/by_level/Grade_03-04/2007 Q17 · always_eight]

5. D — The empty area has 6 squares; all pieces do too, so look at the shape. Turn piece D and it fits exactly; each of the others has one square in a different place. [set-03/by_level/Grade_03-04/2003 Q18 · one_piece]

6. D (27) — The forks have 3, 3, 3 branches; every choice can go with every other: 3 × 3 × 3 = 27. [set-03/by_level/Grade_03-04/2007 Q1 · how_many_routes]

7. E (54) — The box holds 5 × 4 × 3 = 60 cubes. 6 are inside, so 60 – 6 = 54 more. [set-03/by_level/Grade_03-04/2007 Q14 · long_box]

8. E — Only this piece is found in the line. The others don't fit — A: it has the right shapes but in the wrong order; B: its flat part between the two features has the wrong length; C: its flat part between the two features has the wrong length; D: its flat part between the two features has the wrong length. [set-03/by_level/Grade_03-04/2002 Q7 · can_belong]

9. B (30 cm) — The two new edges (2 cm and 2 cm) replace the two removed pieces of the outer edge, which were also 2 cm and 2 cm. So the perimeter is the same as the rectangle's: 2 × (10 + 5) = 30 cm. [set-03/by_level/Grade_03-04/2002 Q6 · corner_cut]

10. A (8) — One bowling ball = 2 footballs = 2 × 2 = 4 tennis balls; 2 bowling balls = 2 × 4 = 8 tennis balls. [set-03/by_level/Grade_03-04/2002 Q9 · chain]

11. D (3) — A drawing is wrong if part of the pole edge is bare or if it reaches more than 4 squares from the pole. Wrong: C, D. So 3 can be right. [set-03/by_level/Grade_03-04/2002 Q15 · count_possible]

12. C (Aisha plays the guitar) — The only possible matching: Aisha – guitar; Chloe – drum; Omar – violin; Ben – xylophone; Nora – harp. [set-03/by_level/Grade_03-04/2002 Q18 · which_true]

13. A — Look where the lines of the picture reach the grey gap: top: 1 line, right: no line, bottom: 2 lines, left: 1 line. Only square A has its lines in exactly those places in this position. The others are mirrored and turned, mirrored and upside down, turned. [set-03/by_level/Grade_03-04/2002 Q1 · turned_square]

14. B (127) — Even one-digit numbers: 4; even two-digit numbers: 45; even three-digit numbers: 11. 4 × 1 + 45 × 2 + 11 × 3 = 127. [set-03/by_level/Grade_03-04/2007 Q23 · even_only]

15. E (48) — The hole is 4 × 4 = 16 cubes in the front and goes 3 cubes back: 16 × 3 = 48. [set-03/by_level/Grade_03-04/2002 Q20 · fill_the_hole]

16. C (8) — WINDOW uses 5 different letters and RAINBOW uses 7. I, N, O and W are in both, so count them once: 5 + 7 – 4 = 8. [set-03/by_level/Grade_03-04/2007 Q3 · letter_cards]

17. A — Two copies have 12 squares, so the rectangle is 1 × 12 or 2 × 6 or 3 × 4. Only two copies of piece A fit together like that. [set-03/by_level/Grade_03-04/2007 Q8 · two_copies]

18. B (48) — Think of Grace and Emma as one block: the block and the other 3 make 4 things to order, 4 × 3 × 2 × 1 = 24 ways. Inside the block Grace and Emma can swap, so 2 × 24 = 48. [set-03/by_level/Grade_03-04/2002 Q22 · side_by_side]

19. D (108) — (1 + 2) × 4 × 9 = 108. Multiplying by 1 changes nothing, so it is better to add the 1 to the smallest other card first. [set-03/by_level/Grade_03-04/2002 Q21 · biggest]

20. A (32) — Each length loses two red cubes at each end, so it drops by 4: 4 × 4 × 2 = 32. [set-03/by_level/Grade_03-04/2003 Q9 · thick_wall]